

FRIDAY, SEPTEMBER 11TH, 2026

CROSS-DOMAIN SYNTHESIS

The Shape of the Worst Case

Past the edge of the record, a single parameter decides everything — and a century of data barely pins it down.

3,059 words · 16-minute read

On the night of 31 January 1953, a deep depression tracking down the North Sea pushed a surge of water against a spring tide and drove it into the funnel between England and the Low Countries. The sea came over and through the Dutch defences in sixty-seven places. By morning roughly 200,000 hectares of the south-west were under salt water, 1,835 people were dead in the Netherlands and around three hundred more in eastern England, and the damage amounted to something near a tenth of Dutch national product.

The government convened a Delta Commission, and the Commission asked a question that sounds administrative and is in fact one of the hardest in applied mathematics. How high should the dikes be?

Not “higher.” A number. And not a number that would have held on the night of the flood, because a defence built to the last disaster is a defence built one disaster short. The Commission wanted a level so high that the sea would reach it, on average, once in ten thousand years. The tide gauge at Hoek van Holland had been recording since 1887. They had, at that point, about two-thirds of a century of readings and a requirement to name a water level with a return period a hundred and fifty times longer than anything they had watched.

That is the general shape of a question which recurs everywhere. What is the strongest earthquake this fault can produce? What is the largest claim this reinsurance layer will see? Each asks for a quantity beyond the largest ever observed, from data that by construction does not contain it.

Why There Are Only Three Answers

The mathematical content of the question was settled, remarkably, before anyone had a flood to apply it to. In 1928 R. A. Fisher and L. H. C. Tippett published a short paper in the *Proceedings of the Cambridge Philosophical Society* on the limiting behaviour of the largest member of a sample [1]. Tippett’s problem was not water at all; he worked at the Shirley Institute, the British cotton industry’s research laboratory, on why yarn breaks, which is a question about the weakest fibre in a length rather than the strongest wave in a winter.

Start with the obvious. If $X_1, \dots, X_n$ are independent draws from a distribution F , then the maximum $M_n = \max_i X_i$ has distribution

$$\Pr(M_n \leq z) = F(z)^n.$$

That is exact and useless. Raising a number below one to a large power sends it to zero everywhere except at the very top of the range, so M_n piles up against the upper end of F and the limit is degenerate. The trick — the same one the central limit theorem plays on sums — is to rescale first: ask for constants $a_n > 0$ and b_n such that $(M_n - b_n)/a_n$ settles down to a non-degenerate law G .

The argument is then almost purely algebraic. The maximum of nk observations is the maximum of k separate maxima of n observations each, so if the rescaled maximum has a limit at all, that limit must reproduce itself under its own maximisation, up to a shift and a stretch:

$$G(z)^k = G(a_k z + b_k) \quad \text{for every } k.$$

This is called max-stability, and it is a strong enough constraint to pin the answer down completely. Fisher and Tippett found that only three families satisfy it. Gnedenko supplied the rigorous proof, together with the conditions under which a given F falls into each family, in a 1943 paper in the *Annals of Mathematics* [2]. Von Mises in 1936 [3] and Jenkinson in 1955 [4] then observed that the three families are one family in disguise. Writing ξ for a single shape parameter, μ for location and $\sigma > 0$ for scale, the generalised extreme value distribution is

$$G(z) = \exp \left\{ - \left[1 + \xi \frac{z - \mu}{\sigma} \right]^{-1/\xi} \right\},$$

defined where the bracket is positive, and read at $\xi = 0$ as its own limit, $\exp\{-\exp[-(z - \mu)/\sigma]\}$.

The parallel with the central limit theorem is worth stating plainly, because it is the reason any of this transfers between fields. The central limit theorem says that sums forget their parent distribution entirely: whatever you added up, you get a Gaussian, characterised by two numbers. Extreme value theory says that maxima also forget their parent — but not entirely. They remember exactly one thing about it, which is the behaviour of its extreme tail, and that memory is compressed into ξ . Sea levels, wind speeds, insurance claims and the breaking strength of glass rods have nothing physical in common. Their maxima are nevertheless described by the same three-parameter law, and differ only in where they sit, how spread out they are, and one number that says what kind of tail they came from.

The Shape Parameter Is the Whole Question

That one number carries all the weight, and it is worth being precise about what it decides.

When $\xi < 0$ the bracket in the GEV vanishes at a finite value, so the distribution has a hard upper endpoint at $\mu - \sigma/\xi$: a largest possible value that no amount of further sampling will exceed. This is the Weibull case, and it is what anything with a physical ceiling produces.

When $\xi = 0$ the tail decays exponentially. This is the Gumbel case, and it has an elegant property: making an event ten times rarer always costs the same $\sigma \ln 10 = 2.303 \sigma$ of height, whether you are going from one-in-ten to one-in-a-hundred or from one-in-a-thousand to one-in-ten-thousand. Dutch engineers call that increment the decimation height, and it is why a Gumbel fit plots as a straight line against log return period.

When $\xi > 0$ the tail is a power law with index $1/\xi$, and moments above that order do not exist. This is the Fréchet case: at $\xi = 0.5$ the variance is infinite, at $\xi = 0.25$ the kurtosis is.

The practical statement of all this is the return level, the value z_T exceeded on average once every T blocks. Inverting the GEV,

$$z_T = \mu + \frac{\sigma}{\xi} \left[\left(-\ln \left(1 - \frac{1}{T} \right) \right)^{-\xi} - 1 \right],$$

with the $\xi = 0$ reading $z_T = \mu - \sigma \ln(-\ln(1 - 1/T))$. And here is the difficulty, stated as arithmetic. Take three laws with the same scale, and choose their locations so that all three agree exactly on the 100-year level — roughly the longest return period a century of annual maxima can constrain directly. Then ask each for the 10,000-year level.

Shape ξ	100-year level	10,000-year level	Extra height
−0.20	0.00	1.20	1.20
−0.10	0.00	2.33	2.33
0.00	0.00	4.61	4.61
+0.10	0.00	9.28	9.28
+0.20	0.00	19.00	19.00

All levels are in units of the scale σ . The last column is the extra height a designer must buy to move from a hundred-year defence to a ten-thousand-year one, and it varies by a factor of nearly sixteen across a range of shapes that a century of data cannot reliably separate. A shape of **−0.2** says the job is nearly done already, because the distribution is running into its own ceiling at **5** σ . A shape of **+0.2** says the job has barely begun.

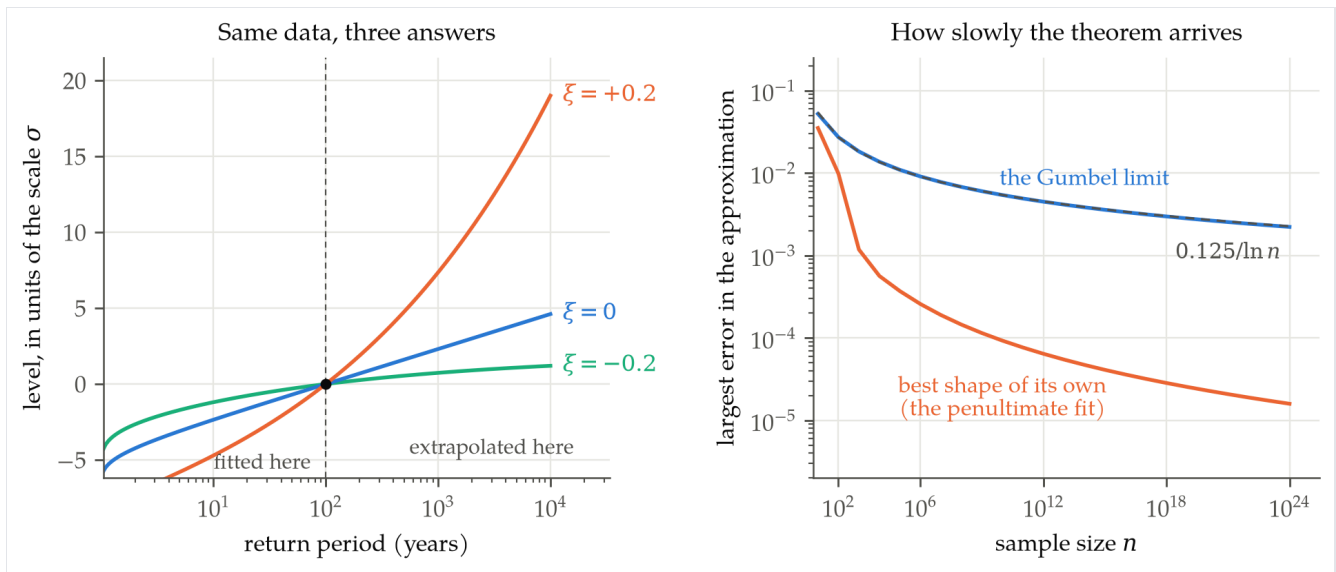


FIGURE Left: three generalised extreme value laws given the same scale and anchored to the same 100-year level, plotted against return period. Inside the fitted range they are hard to tell apart; beyond it they agree on nothing. Right: the largest error in approximating the exact distribution of the maximum of n standard normals, by the Gumbel limit that the theorem guarantees and by the best-fitting law of the same family with a free shape.

A CENTURY OF ANNUAL MAXIMA, AND A DESIGN LIFE OF TEN THOUSAND YEARS. WHAT IS ACTUALLY DOING THE WORK?

Not the data. The data fixes μ and σ well and ξ badly, because the only observations that carry information about the shape of a tail are the handful of largest ones, and a hundred years of records contains a hundred annual maxima of which perhaps five sit high enough to say anything. The extrapolation is carried almost entirely by the assumption about ξ — which is why the honest version of a design standard reports the shape it assumed, and the dishonest version reports the return level alone.

Van Dantzig, asked by the Delta Commission to put the dike problem on a rational footing, did the honest thing. His 1956 paper in *Econometrica* set the question up as a cost-benefit trade-off: raising a dike costs money now, failing to raise it costs an expected loss later, and the optimum equates the marginal cost of height against the marginal reduction in expected damage [5]. The statistical input was an exponential fit to the observed exceedance frequencies of high water — which is to say, a decision that $\xi = 0$. Dutch practice inherited it, and the dike-ring standards that followed ranged from 1 in 250 per year for small areas along the Meuse to 1 in 10,000 for the densely populated rings of North and South Holland. The Commission had first considered a uniform 1-in-125,000 and found it unaffordable, which is the other half of every extreme value problem: the tail asks a statistical question, and the budget answers it.

Crossing the Threshold

Fitting a distribution to annual maxima throws away almost everything. A year in which the three highest surges all exceeded the previous decade's record contributes exactly one number, and a year whose maximum was unremarkable contributes one number of equal weight. For a phenomenon with a hundred years of hourly readings, that is a spectacular waste.

The repair came in the mid-1970s from two directions at once. Pickands, in the *Annals of Statistics* in 1975 [6], and Balkema and de Haan, in the *Annals of Probability* in 1974 [7], showed that if the block maxima of a distribution converge to a GEV with shape ξ , then the excesses over a high threshold u converge, as u rises towards the upper endpoint, to a generalised Pareto distribution

$$H(y) = 1 - \left(1 + \frac{\xi y}{\tilde{\sigma}}\right)^{-1/\xi}, \quad y > 0,$$

with the *same* ξ . That shared parameter is the structural heart of the subject. Two apparently different questions — how large is the yearly maximum, and how far past a high bar does a crossing go — have the same answer, and the answer is a number about the parent's tail rather than about either procedure.

Peaks-over-threshold analysis uses every exceedance rather than one per block, typically multiplying the usable sample several-fold. It buys that with a new problem: choosing u . Set the threshold too low and the asymptotic result does not yet hold, so the estimate is biased; set it too high and too few points remain, so it is noisy. There is no formula for the right threshold, only diagnostics. Bruce Hill's 1975 estimator, published in the same volume of the *Annals* as Pickands' paper, computes a tail index from the top k order statistics alone and makes the trade-off visible: plot it against k and look for a plateau [8]. Practitioners have spent fifty years arguing about where the plateau is.

One Parameter, Four Literatures

What makes this worth learning once rather than four times is that the same ξ shows up wherever anyone asks about a worst case, and its sign tells you immediately what kind of world you are in.

Brittle materials. A ceramic component fails at its largest flaw, so its strength is the *minimum* over many competing weaknesses — an extreme value problem with the sign flipped. Waldemar Weibull, in two 1939 memoirs for the Royal Swedish Institute for Engineering Research and then in a 1951 paper whose title claimed, accurately and immodestly, wide applicability [9], [10], derived the survival law that carries his name from exactly this weakest-link argument, with Griffith's 1921 account of fracture at pre-existing flaws supplying the physics underneath [11]. The Weibull modulus m is the shape parameter in disguise: $\xi = -1/m$. Its consequence is the size effect, and it is brutal. Strength scales as $V^{-1/m}$, so for a typical structural ceramic with $m = 6$, doubling the volume costs 11 per cent of the strength and a thousandfold increase in volume leaves 31.6 per cent of it. This is why the strength of a laboratory coupon is not the strength of the part, and why ceramic design specifies a survival probability rather than a strength.

Fire insurance. Alexander McNeil fitted the generalised Pareto to Danish industrial fire losses from the 1980s and obtained $\xi = 0.684$ above a threshold of 20 million kroner [12]. Follow the arithmetic: the tail index is $1/0.684 = 1.46$, comfortably below 2, so the variance of a Danish fire loss does not exist. Sidney Resnick, discussing the same data in the same volume, made the point that the usual asymptotics assuming a finite second moment simply do not apply [13]. An insurer who prices an excess-of-loss layer from a sample standard deviation is computing a statistic that has no population counterpart and will drift upward forever as more data arrives.

Financial returns. Gopikrishnan and colleagues, working with tens of millions of observations, found the cumulative distribution of index returns decaying as a power law with exponent 3.1 ± 0.03 in the positive tail and 2.84 ± 0.12 in the negative one [14]. In shape-parameter terms that is $\xi \approx 0.32$ and 0.35 : the variance exists comfortably, but the fourth moment does not. Sample kurtosis for daily returns is therefore a number that never settles, and two analysts with different windows will disagree permanently while both are right about their own data.

Human lifespan. Einmahl, Einmahl and de Haan applied the machinery to the ages at death, recorded in days, of roughly 285,000 Dutch residents who died between 1986 and 2015 at age 92 or older [15]. They found $\xi < 0$ — a finite endpoint — and estimated it at 115.7 years for women and 114.1 for men, with no trend across the thirty years despite the number of people reaching 95 nearly tripling over the same period. The oldest woman in their sample reached 115.2, so the estimated ceiling sits half a year above the highest observation, which is what a sharply negative shape does. The honest complication is that Jeanne Calment died at 122.45 years, well outside that endpoint. She was French, not Dutch, and one of a handful of verified cases worldwide; whether she refutes the estimate or merely lies outside the population it was fitted to is precisely the kind of question the theory cannot settle from within.

Field	Fitted shape ξ	Tail index $1/\xi$	What it implies
Ceramic strength, $m = 6$	-0.167	—	hard upper limit; strength falls with size
Dutch lifespan, women	< 0	—	endpoint estimated at 115.7 years
Equity returns, lower tail	$+0.352$	2.84	no kurtosis; sample kurtosis never converges
Danish fire losses	$+0.684$	1.46	no variance; sample standard deviation is meaningless

What the Theorem Does Not Promise

It would be a poor advertisement for the subject to leave it there, because the three most common ways of misusing extreme value theory are all failures to read the small print.

The first is the assumption that the limit has arrived. Fisher and Tippet noticed, in the founding paper itself, that convergence to the Gumbel law is agonisingly slow when the parent is Gaussian. Peter Hall made it precise in 1979: with optimal norming constants, the error decays like $1/\ln n$ [16]. That rate is worse than it looks. Working with the exact distribution of the maximum of n standard normals, the

largest discrepancy from the Gumbel limit is 0.0091 at $n = 10^6$ and 0.0045 at $n = 10^{12}$ — squaring the sample size halves the error, so a trillion observations are twice as good as a million and no better. And the remedy Fisher and Tippett proposed is stranger still. If instead of the correct limiting shape $\xi = 0$ you fit the best shape of the same family, you get $\xi = -0.038$ at $n = 10^6$ and an error of 0.00026, thirty-five times smaller. The exactly right asymptotic answer is beaten, by more than an order of magnitude, by a deliberately wrong one. This is the penultimate approximation, and it is a useful corrective to the idea that a limit theorem is a licence.

The second is independence. Storms last for days; hot spells cluster; a market in distress produces its extreme days consecutively. Leadbetter's framework for stationary dependent sequences introduces an extremal index $\theta \in (0, 1]$, and its content is that a dependent series behaves in its extremes like an independent one with $n\theta$ observations instead of n [17]. The reciprocal $1/\theta$ is the mean size of a cluster of exceedances. Threshold methods that treat every exceedance as a fresh event on a clustered series count the same storm several times, and are correspondingly overconfident.

The third, and now the most consequential, is stationarity. The entire apparatus assumes the underlying distribution is not moving. In June 2021 a heat dome over the Pacific Northwest produced 49.6 °C at Lytton, British Columbia — 4.6 °C above the previous Canadian national record, in a variable where records normally fall by tenths. A rapid attribution study fitted a GEV including the event and concluded it was virtually impossible in a pre-industrial climate [18]. The IPCC's sixth assessment puts the general version in terms the machinery recognises: a daily heat extreme that occurred once a decade in the 1850–1900 climate is now roughly three times as frequent, and is projected to become about four times as frequent at 1.5 °C of warming and about nine times at 4 °C [19]. A return period is a statement about a distribution, and a moving distribution makes the phrase “hundred-year event” an assertion about which century you mean.

Even in a stationary world that phrase misleads. A one-in-a-hundred-year flood has a 26.0 per cent chance of occurring at least once in a thirty-year mortgage and a 50.5 per cent chance across a seventy-year life. And the Dutch standard that started this — one in ten thousand per year — carries very nearly a 1 per cent chance of being exceeded in any given century. A design standard is not a promise that something will not happen. It is a statement about how much probability a society has decided to buy.

The Netherlands, characteristically, has kept revising the purchase. The Maeslant barrier that now guards the waterway at Hoek van Holland is built for a surge of five metres above datum. Since 2017 the Dutch Water Act has replaced the old uniform dike-ring exceedance standards with risk-based failure probabilities set segment by segment, ranging from 1 in 300 to 1 in a million per year, calibrated so that an individual's annual probability of dying in a flood stays below one in a hundred thousand; the framework grew out of a line of optimisation work running from van Dantzig's original formulation to the modern treatment by Eijgenraam and colleagues [20]. That is a different kind of answer from the one the Delta Commission gave in the 1950s, and a better one. But it rests on the same fragile object: a shape parameter, estimated from the few largest numbers anyone has written down, standing in for every wave that has not yet arrived.

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